Technology And Globalisation: How Social Media (Narrow Artificial Intelligence) Algorithms and Large Language Models by Internet Companies Have Replaced the Main Stream Media as Agenda Setters For the Global Culture.

Moses Sichach

Mirrlees (2020) defines global culture first as an individual cosmopolitant feeling or affective identification with a global community. He also adds another dimension to the definition by describing it as a particular social structures, characters and practices of the hegemony that have spread across boarders.

Cuture deals with the dissemination of meaning i.e how we make meaning. Language, music and images constitute major forms of symbolic expression. (Manfred, 2003).

Since the golden age of television in the 1950 to atleast 2004 with the advent of Facebook, mainstream media has been the primary way in which music, languages and images were transmitted globally, it was the primary vehicle for the spread of western culture (mordernisation) which later formed the cosmopolitant global culture. Purvis (2006) states that the media advocates for social concerns and facilitates communication and exchange of positive cultural values among different societies.The media plays a critical role in disseminating societal norms, ideologies, and customs. Because of the media, socialization is now possible and much easier. (Ivy Panda, 2023). Mainstream media does this by determining which issues are important and giving them coverage, this is known as agenda setting.

The beginning of agenda setting theory can be traced as far as 1922, when Walter Lippmann expresses his concern on the vital role that mass media can do in influencing

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> the setting of certain image on the public‟s mind (Lippmann, 1922: 9-16). In portraying the influence of mass media, Lippmann gives an example of individuals who supposed to be enemies while their countries are at war. Instead of becoming enemies, without having access to information about the war through media, those individuals are able to live harmoniously in a secluded island. Lippmann indicates on how mass media can set a particular agenda which can influence the opinions of the public. However, he never uses the term „agenda setting theory‟ in his book. Nevertheless, he did generate the foundation for the agenda setting theory. (Mohd Zain, 2014)

According to agenda setting theory, the media is the primary entity that selects and prioritizes which news stories to report and prioritize based on what they believe people will be interested in. People's perceptions of cultural, social, and political news can be influenced by how the media filters and shapes what they see. People have access to information that they believe is most important to society and will have the most influence over them thanks to the media. (indeed.com, 2022).

Peoples perceptions in this case are impacted through the mere exposure effect, is a psychological phenomena in which people develop a preference for objects or persons with whom they are more familiar than others. Repeated exposure leads to increased familiarity. As a result, this phenomenon is also known as the familiarity effect. (Falkenbach et al , 2009)

Rouse (2023) defines Artifiacial Narrow Intelligence as a sort of artificial intelligence in which a learning algorithm is designed to perform a single task and any information gained from doing so is not automatically applied to other tasks. A large language model is a type of artificial intelligence (AI) program that understands, summarizes, generates, and predicts new content

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using deep learning techniques and enormously big data sets. The phrase generative AI is also strongly associated with LLMs, which are a sort of generative AI that has been specifically designed to assist in the generation of text-based material. (Kerner, 2023)

Artificial Narrow Intelligences employed by Social Media Companies as alorithims to serve up optimal content for viewers and advertisers and the Intergration of Large Language models such as Chat GPT and Google Bard to browsers are exposing the global audience to content that fits their terms and conditions, are advertiser friendly and usually follow legal requirements of western countries as they are incorporated there. The result is pushing of specific Global narratives such as Climate change, gender vs and sexuality, Western heroes vs eastern and African villains.

Statista.com (2023) has tracked that the average user spends 151 minutes daily on social media sites. According to a research by New York-based agency We Are Social and social media management business Hootsuite, the typical Internet user aged 16 to 64 spends six hours and 58 minutes each day online across all internet platforms. This is for the 5.3 billion users with access to the internet globally with the number expected to rise to 6.5 billion by 2025 (Shewale 2023)

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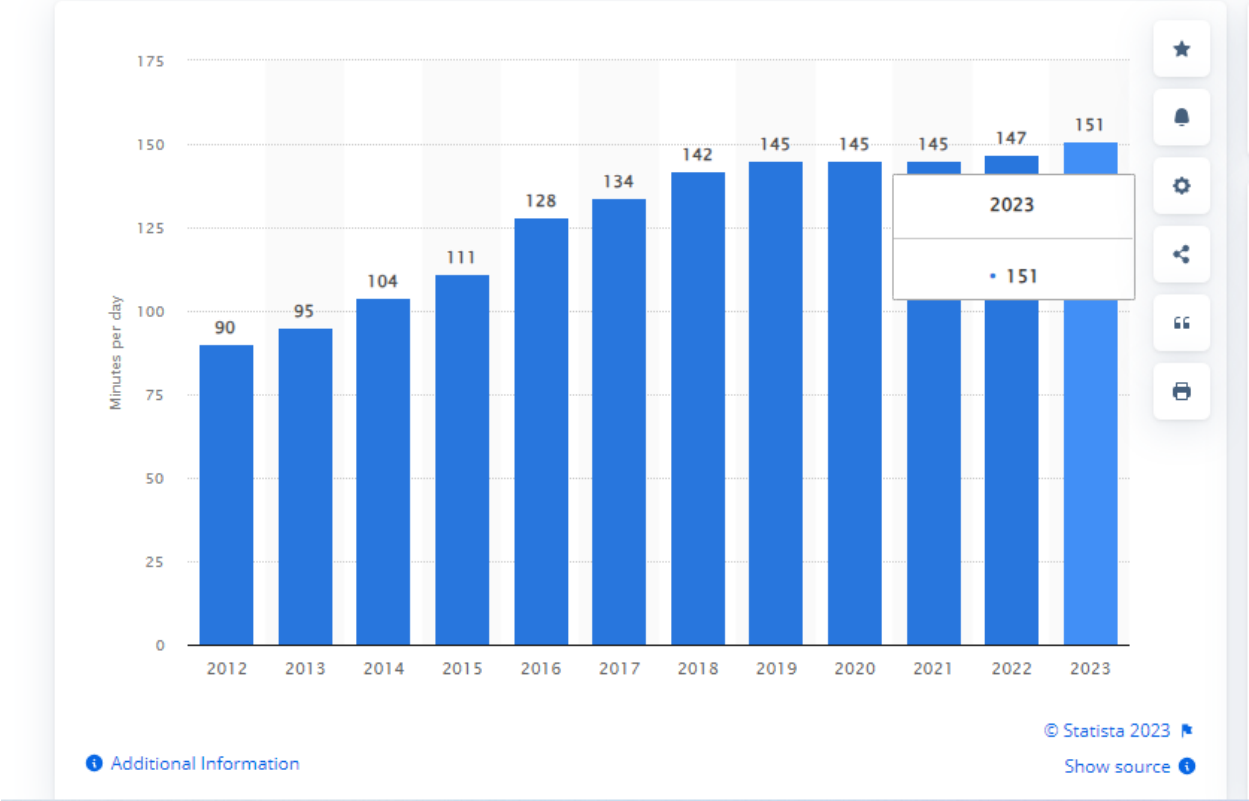

Figure 1. Daily time spent on social networking by internet users worldwide from 2012 to 2023 (Statista.com, 2023)

This is in contrast to the average of 2hrs 33 minutes spent on traditional TV stations for a population of 5.4 billion. This number includes streaming sites such as netflix, disney plus, hulu and apple TV (TV scientific 2023). A far cry from the afore mentioned 6 hrs and 58 minutes on the internet,

This paper looks to highlight the forced shift from the main stream media as an agenda setting machine for the global culture towards sillicon valley mutinationals (Google, Microsoft, Meta, X corp (Previously twitter) and TikTok) with the help of Narrow Arificial Intelligence(Algorithms) and Large Language Models through what is trending and promoted to a large audience. This

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shift is significant because it represents a departure from the traditional paradigms of agenda setting that were once shaped by established media conglomerates. Technology companies have used sophisticated algorithms and language models to harness the power of data-driven insights to not only identify but also actively shape and amplify trends and by extension the shared global cultrure as a result.

Ross et al (2022) digs at the phenomena of echo chambers, which are maintained by limited social media algorithms. According to the study, algorithmic content curation frequently reinforces users' pre-existing ideas by displaying information that matches their tastes. This selective exposure unintentionally promotes dominant cultural narratives, ultimately amplifying Western perspectives.

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Figure 2 : An Image generated by MID JOURNEY (AI Image model) depicting a white man robbing a store.

Sun et al. (2018) expand on this by emphasising the socio-cultural ramifications of echo chambers, emphasising the detrimental influence on pluralism and diversity of media consumption. According to the study, such algorithms generate information silos, limiting exposure to diverse perspectives and supporting the Western-dominated narrative.

González-Bailón et al. (2019) provide a complementary viewpoint by investigating the role of algorithmic content curation in determining global news dissemination. The study found that because Western news sources are so prevalent, algorithms tend to prioritise Western perspectives in the global information landscape, thereby excluding narratives from non-Western civilizations.

Shifting focus to Large Language Models, Johnson and Martinez (2019) contribute to the discourse by highlighting the biases embedded in these models. They argue that the reliance on English-language training data disproportionately favors Western terminology and concepts, leading to biased text generation. This linguistic bias inadvertently promotes Western cultural narratives and reinforces linguistic hegemony.

The integration of Large Language Models (LLMs) and algorithms as agenda setters poses several challenges that have implications for information diversity, cultural representation, and societal discourse. Drawing insights from the paper, the following challenges emerge:

1. Homogenization of Information:

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The utilization of LLMs and algorithms to set the agenda could lead to the homogenization of information. These technologies are designed to optimize user engagement by presenting content that aligns with users' preferences. As a result, individuals may be exposed to a limited range of perspectives and narratives that match their pre-existing beliefs. This could contribute to echo chambers, where individuals are isolated from diverse viewpoints and alternative cultural narratives, hindering a holistic understanding of global issues.

2. Reinforcement of Dominant Narratives:

LLMs and algorithms are susceptible to biases present in the data they are trained on. If these biases are not actively addressed, these technologies might inadvertently amplify and reinforce existing dominant cultural narratives. This perpetuation of one-sided viewpoints can hinder the recognition and appreciation of diverse cultural voices and experiences.

3. Marginalization of Non-Western Narratives:

Given the prominence of Western data sources and language in training datasets, LLMs and algorithms could inadvertently marginalize non-Western narratives. This can lead to the underrepresentation or misrepresentation of cultural experiences and perspectives from regions outside the Western world. The result is an imbalance that perpetuates cultural hegemony.

4. Loss of Cultural Diversity:

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As LLMs and algorithms prioritize content based on user preferences and engagement metrics, there is a risk of overlooking culturally significant content that might not align with mainstream interests. This could lead to the loss of cultural diversity and heritage, as content that is not deemed popular or trending might be suppressed.

5. Polarization and Fragmentation:

The selective exposure facilitated by algorithmic content curation can contribute to increased polarization and fragmentation within societies. Users may gravitate towards information that confirms their existing beliefs, reinforcing divisions and inhibiting constructive dialogue across cultural and ideological boundaries.

6. Erosion of Traditional Media Influence:

The shift from traditional media to technology companies as agenda setters could erode the influence of established media outlets. This may undermine the role of journalistic ethics, fact-checking, and editorial oversight, leading to the dissemination of unchecked and potentially misleading information.

7. Loss of Human Editorial Judgment:

The automation of agenda setting through algorithms and LLMs reduces the role of human editorial judgment in curating content. This can result in a lack of critical evaluation and context, leading to the propagation of sensationalized or incomplete information.

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In conclusion , this paper has investigated the growth of global cultural diffusion, the role of mainstream media, and the advent of technology-driven agenda setting via the integration of Large Language Models (LLMs) and algorithms. Mirrlees' (2020) concept of global culture as both affective identification with a worldwide community and the expansion of specific social institutions and practises emphasises the intricacies of modern cultural dynamics. Language, music, and images, among other symbolic expressions, impart cultural meaning, affecting how we interpret and participate with the world.

From the post-World War II age of television to the introduction of social media platforms such as Facebook, mainstream media has played a significant part in spreading Western culture and facilitating cosmopolitan global culture. However, with the rise of digital behemoths like Google, Microsoft, and Meta, a fundamental shift is taking place. The ability to control cultural narratives is shifting away from traditional media conglomerates and towards technology firms that use Narrow Artificial Intelligences and LLMs. These systems curate information to meet user preferences, advertiser needs, and legal standards, which are frequently based in Western cultures. As a result, specific global narratives, such as climate change and cultural divides, are being pushed to a wider audience.

This transformation has far-reaching consequences. The issues that come as a result of LLMs and algorithms becoming agenda setters are diverse. The risk is that dominant narratives will be perpetuated, non-Western perspectives will be marginalised, and cultural diversity will be eroded. Echo chambers are becoming more intense, contributing to societal polarisation and division. The decline of traditional media dominance and the loss of human editorial judgement exacerbate these issues.

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It is critical for technology businesses to address biases in algorithms and training data, prioritise cultural diversity and inclusivity, and promote openness in content curation as they navigate these problems. Individuals must have media literacy and critical thinking abilities in order to engage critically with the information offered to them. As technology continues to transform the landscape of cultural diffusion, constant study and ethical considerations are essential.

Finally, the convergence of global culture, media, and technology highlights the need for a nuanced approach to agenda setting—one that values inclusivity, respects cultural pluralism, and harnesses technology's potential to facilitate a more informed, interconnected, and culturally rich global society.

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